

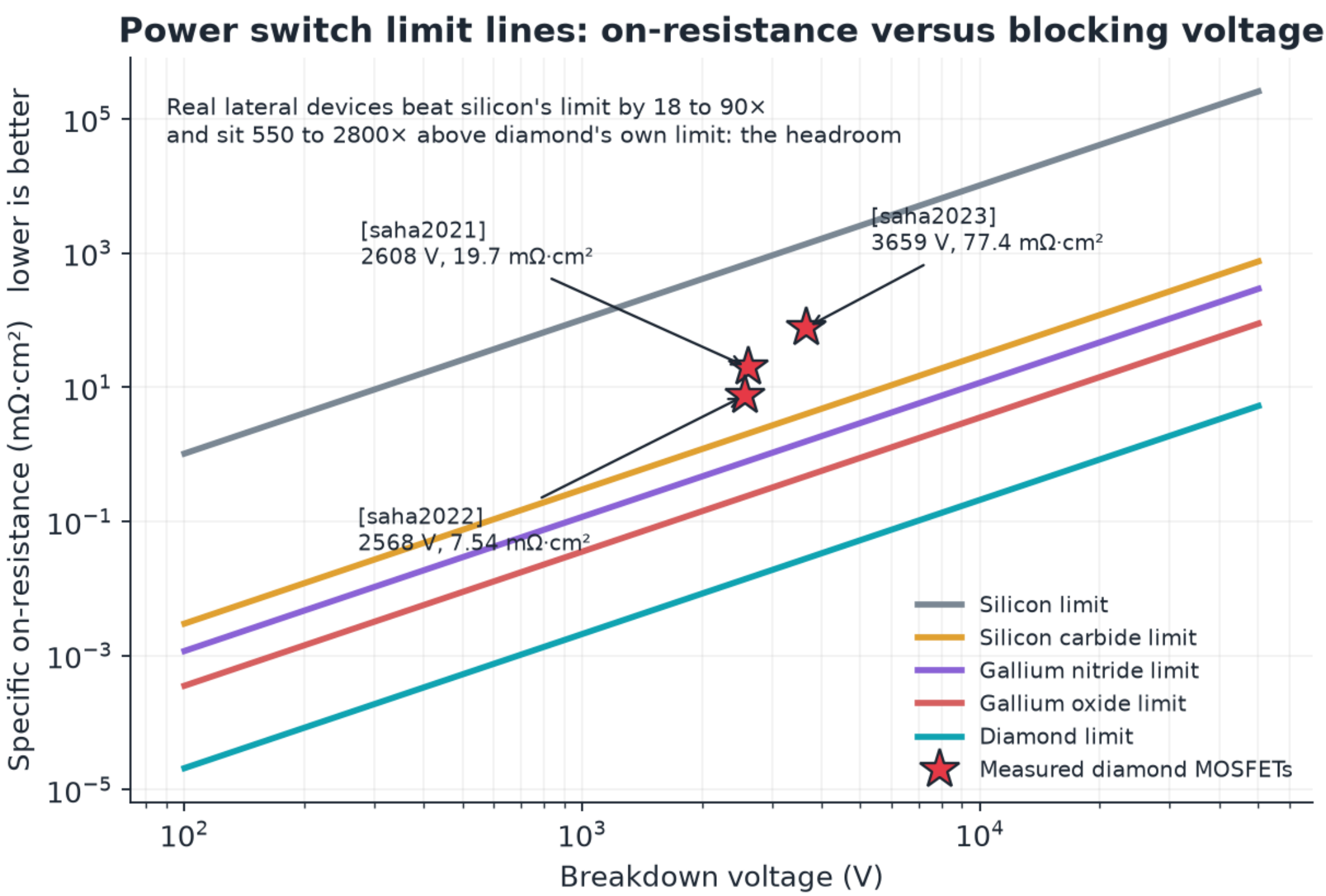
DIAMOND CHIPS, FROM WAFER TO QUBIT

Aleksander Norman · ADAMAS: an open, tested framework for diamond electronics and room-temperature quantum processors

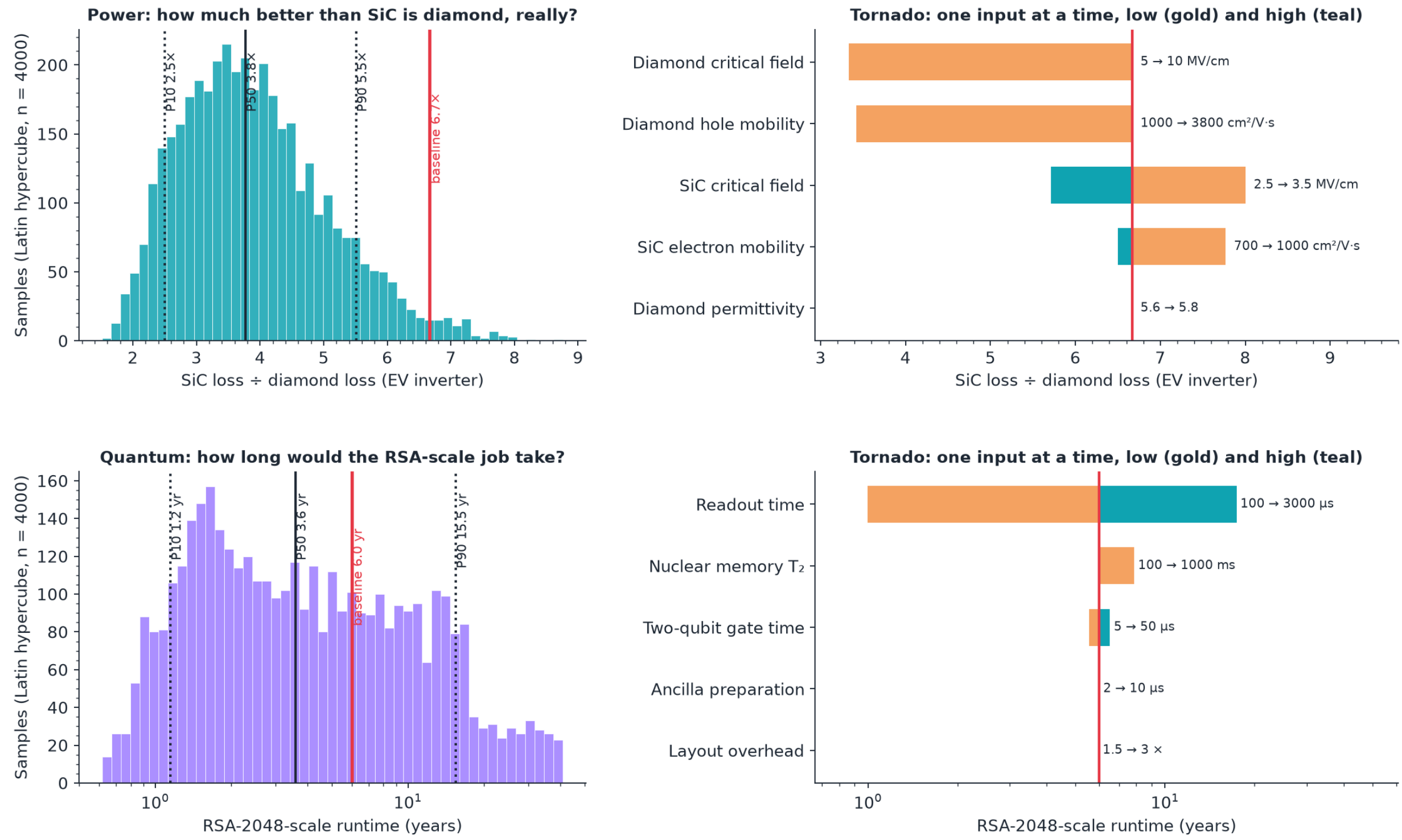
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github.com/Normansrule/adamas-diamond-framework

Diamond has the largest breakdown field and thermal conductivity of any practical semiconductor, and a defect, the nitrogen-vacancy (NV) center, that is a qubit at room temperature. ADAMAS follows one path from wafer to quantum processor with executable, cited models: **541 references**, **52 regenerable figures**, a process traveler, a placed processor, circuit-level error correction, and experiments from US\$100. **Every number on this poster is generated by the package.**

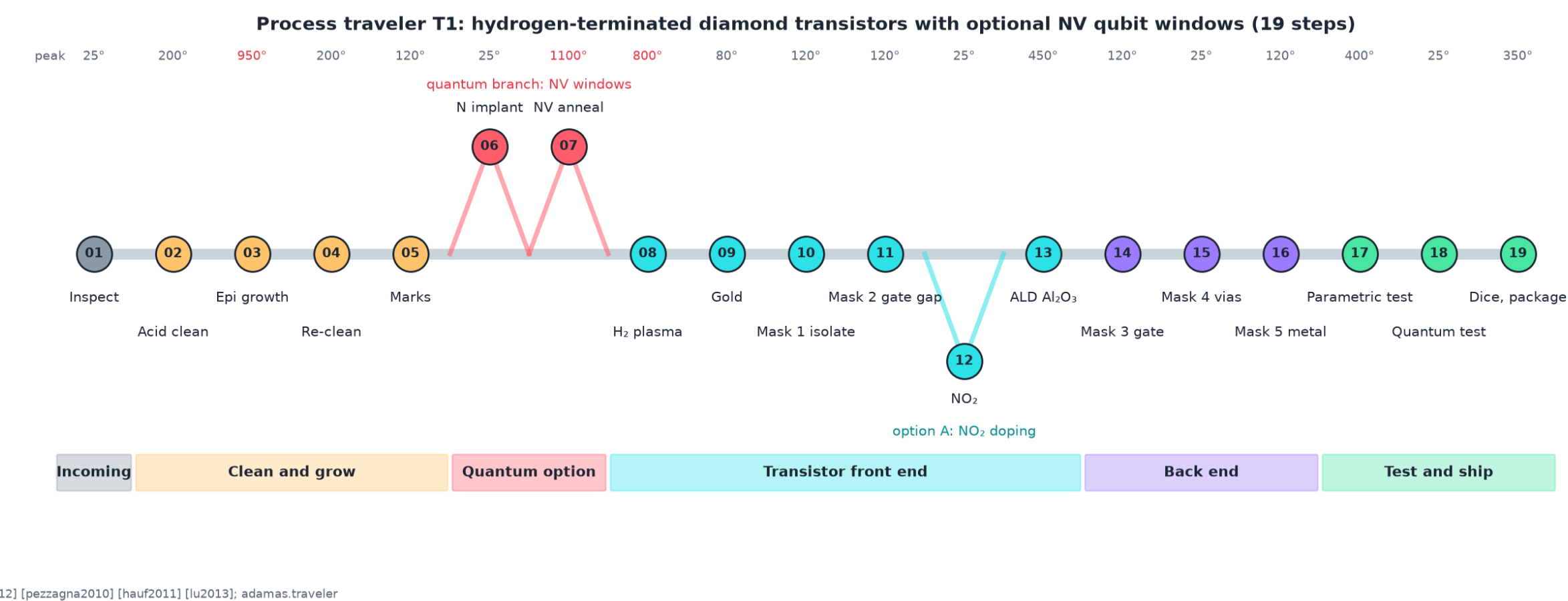
1 · Power: a real but bounded advantage



Measured hydrogen-terminated diamond MOSFETs reach 3659V and beat silicon's limit by **89x**. In an 800 V electric-vehicle inverter an ideal diamond switch loses 3.8W against 29.7W for silicon carbide. Propagating the published spread of diamond's critical field and hole mobility gives a median advantage of **3.8x** (P10 to P90: 2.5 to 5.5x). Sobol indices rank what to measure first: diamond hole mobility, by a Hall measurement costing a few thousand dollars.



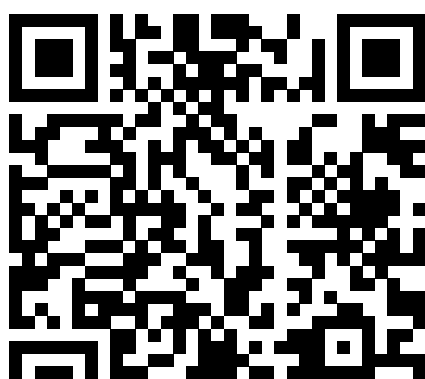
2 · How to build it



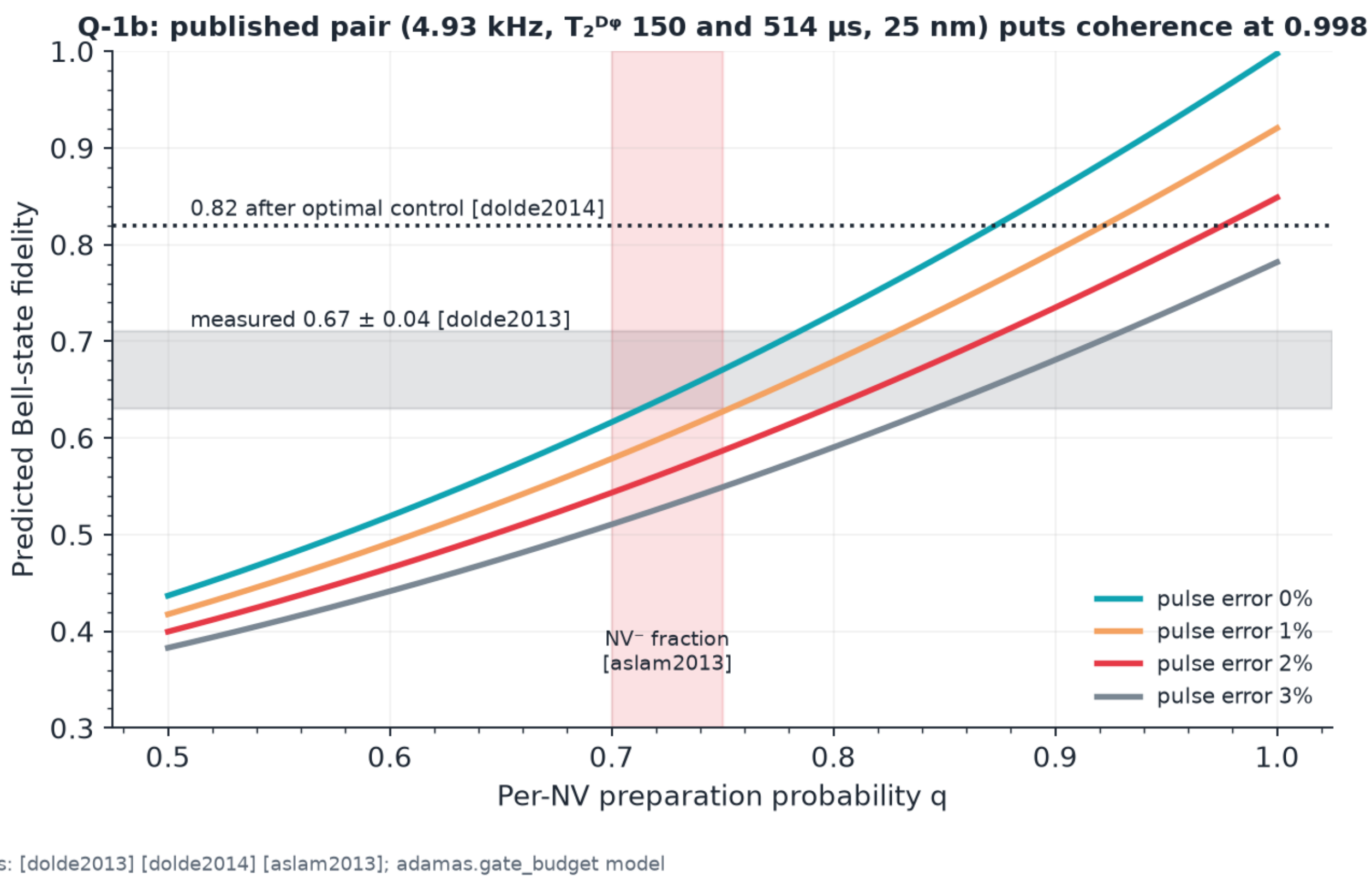
A 19-step traveler: hydrogen termination makes the channel, gold protects it, one oxygen plasma isolates devices, and all heat is spent before the first metal. The PDK-0 kit places the 4-bit DIA-4 processor (323 cells) in about 1 mm².

Validation and reproducibility

19 validation checks (18 pass) against published results, independent tools, and known answers; 500 references verified against Crossref. `pip install -e ".[dev]" && pytest && make figures paper talk poster` regenerates every-

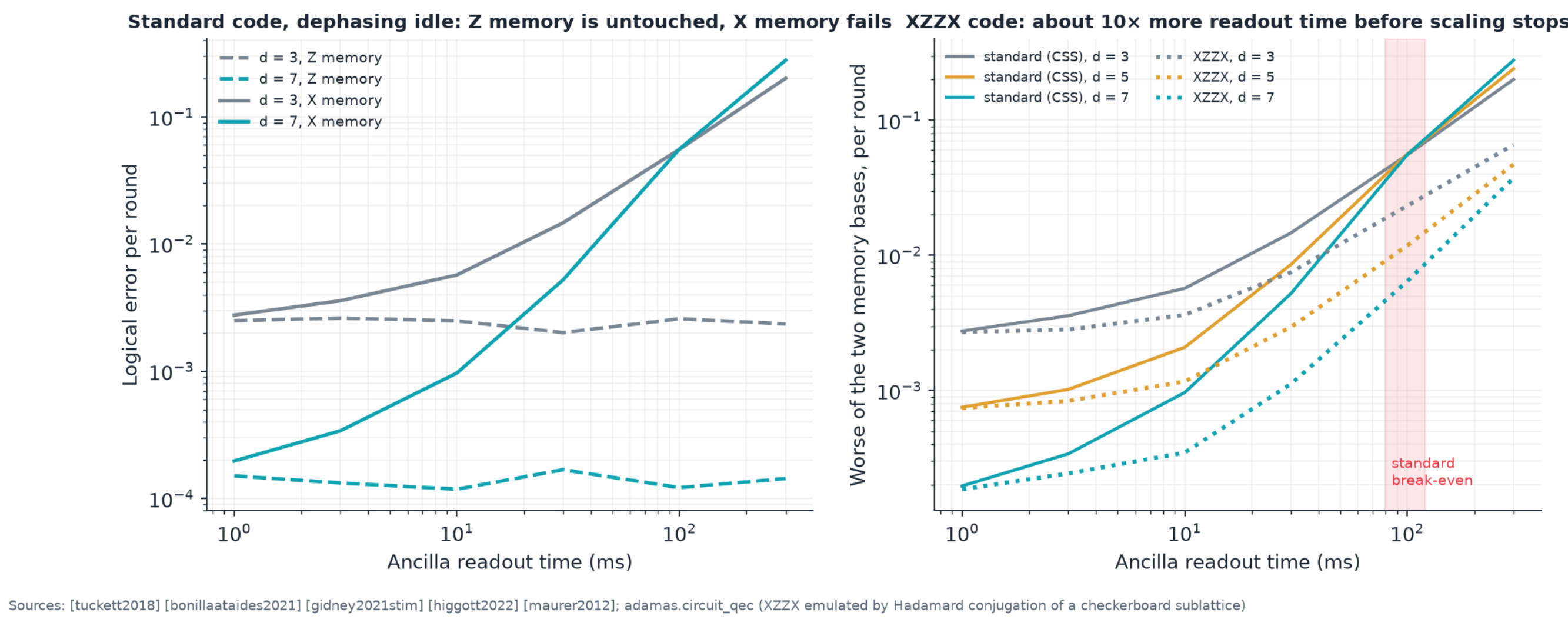


3 · The qubit bottleneck is preparation



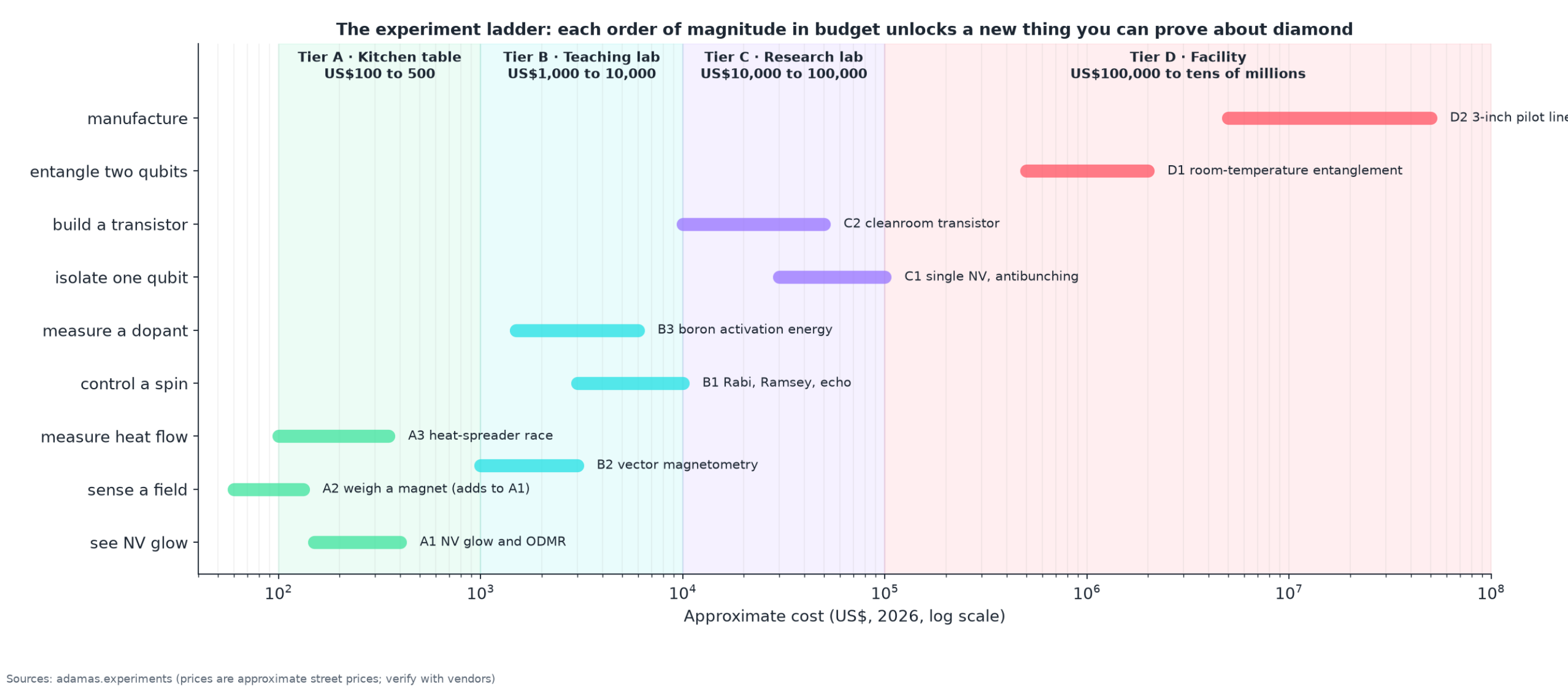
With the published 25 nm pair, coherence alone allows a Bell fidelity of **0.998**; the measured 0.67 means the NV⁻ charge state was ready only 0.75 of the time.

4 · Error correction with slow readout



Circuit-level simulations (Stim, PyMatching): scaling stops near 100 ms readout for a 1 s nuclear memory; the bias-tailored XZZX code extends this to about **1000 ms**. Native XZZX circuits with biased gates raise the threshold to **1.59%** at bias 100, while the standard code drops to 0.63%. An RSA-2048-scale machine fits on a 5.7 mm die but runs **6.0 years** at 1 ms readout (P10 to P90: 1.2 to 15.5). With biased gate noise the standard code needs 399 M qubits; XZZX needs **67 M**.

5 · Try it: experiments from US\$100



Ten experiments in four budgets with parts, steps, safety, and expected results; the A1 ODMR kit ships with Raspberry Pi Pico firmware and a fitter.